

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – MCQ Test

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	Assessment:	Assessment Tool 3 – MCQ Test
Weightage:	30% of CIA	Total Marks:	100
CO3 Statement:	<i>Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, BTrees, Red-Black Trees, and Splay Trees.</i>		
Student Name:	J. MEHER KOUSHIK	Reg. No.:	192525211
Section / Batch:	AI&ML	Date:	12/08/2026

Code of Conduct

I J. MEHER KOUSHIK (192525211) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: J. MEHER KOUSHIK

Instructions

- This test contains 25 MCQ Test questions (4 marks each), Total: 100 Marks.
- When a student answer using MCQ, in evaluation the answer should be with following requirements:
Identify the most appropriate answer, conceptual understanding, problem-solving ability, application of knowledge.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Preliminaries – Tree Terminology	Root, height, level, siblings and sub tree	1–2	8
Binary Tree, Binary Search Tree	Properties, binary tree and binary search tree, insertion and deletion	3–9	28
Tree Traversals	Traversal types, In order, Post order and Pre order	10–14	20
AVL Tree	Balancing factor, Rotation, Types of rotation, examples	15–16	8
Applications of Search Trees	Usage of binary tree	17	4
B Tree - 2-3 tree, 2-3-4 tree	Properties, Example and Operations	18–19	8
TRIE	Properties, structures and Operations	20	4
Red Black Tree	Properties, Balancing and Operations	21–22	8

Splay Tree	Playing, Types and examples for insertion and deletion	23–25	12
GRAND TOTAL:		100 Marks	

MCQ

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	20	Demonstrates thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge	Shows good understanding with minor conceptual gaps	Basic understanding; several misconceptions present	Limited understanding; major concepts misunderstood
Application Skills	20	Correctly applies concepts to solve complex and scenario-based MCQs	Applies concepts correctly in most moderate-level questions	Struggles with application in complex scenarios	Unable to apply concepts effectively
Analytical Thinking	30	Consistently selects correct answers for high-order questions	Performs well in moderate analytical questions	Limited success in analytical questions	Unable to interpret analytical/problem-based questions
Time Management	20	Completes test efficiently with time for review	Completes test within time	Struggles to complete test on time	Unable to complete test
Accuracy of Responses	10	90–100% correct answers	75–89% correct answers	50–74% correct answers	Below 50% correct answers

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO3_AT3_MCQ

← Back

QUESTIONS
Q1 ✓

Tree Traversal
4 marks

Q2 ✓

Binary Search Tree
4 marks

Q3 ✓

Red-Black Tree
4 marks

Q4 ✓

Red-Black Tree
4 marks

Q5 ✓

B-Tree
4 marks

Q6 ✓

Tree Terminology
4 marks

Q1 Tree Traversal

4 marks

You analyze the post-order traversal of a tree representing project tasks: [D, E, B, F, G, C, A]. You want to know

A. 3

B. 4

C. 5

D. 2

Save Answer

Next →

CO3_AT3_MCQ

← Back

QUESTIONS
Q1 ✓

Tree Traversal
4 marks

Q2 ✓

Binary Search Tree
4 marks

Q3 ✓

Red-Black Tree
4 marks

Q4 ✓

Red-Black Tree
4 marks

Q5 ✓

B-Tree
4 marks

Q6 ✓

Tree Terminology
4 marks

Q25 Splay Tree

4 marks

After accessing a node in a Splay Tree, that node is moved to:

A. Leaf

B. Middle

C. Root

D. NULL

Save Answer

My Assessments

Course: CSA0305RA

PSEUDOCODE WRITING TASK OPEN

CO5_AT1_Pseudocode Writing Task

0/5 answered

Attempt →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2) OPEN ✓ Completed

DEBUGGING & OPTIMIZATION TASK

10/10 answered

Review →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1) OPEN ✓ Completed

DEBUGGING & OPTIMIZATION TASK

10/10 answered

Review →

CO3_AT3_MCQ OPEN ✓ Completed

Multiple Choice Questions - Non Linear Data Structures

25/25 answered

Review →

CO1_AT1_C CODE OUTPUT PREDICTION OPEN ✓ Completed

C CODE OUTPUT PREDICTION

10/10 answered

Review →